DIOGO SILVA

Gameplay & AI Engineer | C++ · Unreal Engine 5 | AAA Games

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PROFILE

Gameplay and **AI Engineer** with 3+ years of **AAA** industry experience, shipping titles on **PC** and **Consoles** using **C++** and **Unreal Engine 5**. Core contributor to multiple titles of varying genres, owning various Gameplay and AI systems from melee and ranged combat, to cinematic boss fights and Photo Mode. Backed by an academic background in **Computer Science & Engineering** (MSc, GPA 19/20) and vast personal portfolio spanning game prototypes, machine learning systems and other software.

WORK EXPERIENCE

Gameplay Software Engineer | 2K Games *Jun 2025 – Present*

Titles: *Mafia: The Old Country (shipped)* · *Unannounced Action Title*

- **Shipped Mafia: The Old Country**, identifying and solving critical bugs on different platforms and across multiple systems ahead of release.
- **Supported** Mafia: The Old Country **post-launch** with targeted fixes, improvements and new features such as new races and combat encounters for a free, open-world content update.
- **Led Photo Mode** end-to-end development - scoped constraints, iterated rapidly, and delivered a polished feature in a short timespan. The feature was very well received by the community, being a main post-release player engagement generator, earning accolades and being featured in thephotomode.com.
- Adopted Unreal Engine **AngelScript** to develop gameplay systems and bridge the gap between engineering and tech designers, enabling faster iteration, reducing reliance on C++ for designer-facing systems and preventing dependency on performance-heavy Blueprint-driven game logic.
- Designed and implemented **Recoil** and **Weapon Systems** framework with deep designer tunability via Data Assets, reducing iteration time for the weapons team.
- Created a flexible, data-driven **Inventory System** designed for reuse across future Mafia titles.

Graduate Software Engineer | 2K Games *Jul 2023 – Jun 2025*

Titles: *Mafia: The Old Country* · *Unannounced Online Sports Title*

- **Owned Core Gameplay Features** for Mafia: The Old Country's melee combat and boss fights — including combos, ability transitions, motion warps, and conversation subsystems — from vertical slice to ship.
- Developed **Utility-Based AI Systems** and **Collision Avoidance Algorithms** for an unannounced online sports simulation title.
- Leveraged Unreal Engine's **Gameplay Ability System** (GAS) daily on different game systems.
- Collaborated with designers to implement **Data Asset-Driven Gameplay Systems** and delivered custom **Debugging** and **Asset Management Tools** to facilitate rapid gameplay prototyping.
- Collaborated with two 2K studios (**Hangar13** and **2K SportsLab**), gaining experience working on various AAA game genres under different teams and production cycles. Flown by the company to the US, Canada, and UK on multiple occasions for studio visits and cross-team collaboration. Recognised as one of the highest contributors in the 2K Graduate Program cohort.
- Completed official Unreal Engine 5 training provided by Epic Games.
- Presented feature work to Studio Directors and pitched new game ideas directly to 2K's Executive Team.

SKILLS

Languages: C++, Python, Blueprint, Unreal Engine AngelScript
Engines & Tools: Unreal Engine 5, Unity Engine, Perforce, Git
Gameplay Engineering: Gameplay Ability System, 3Cs, Inventory, Melee Combat, Ranged Weapons
AI Engineering: Pathfinding, GOAP, State Trees, Behaviour Trees, Utility AI, Collision Avoidance
Other: Animation Engineering, Machine Learning, Software Design Patterns, OOP, SCRUM / Agile
Languages (spoken): Portuguese (native), English (fluent - C2-CEFR), Japanese (beginner- N5)

EDUCATION

Master's in Computer Science & Engineering *Sep 2020 – Nov 2022*

University of Lisbon | GPA: 19.0 / 20.0

Highest grade average in the cohort. **Academic Excellence Award** recipient. Specialized in **Intelligent System**s and **Game Development**; electives in Computer Graphics and Virtual Reality. **Published** accompanying thesis research at **IEEE ISM 2022** (Naples, Italy). Worked as a paid **Undergraduate Researcher** solo developing a web platform for the University's Pedagogical Council. Actively participated in multiple **student associations**, including **GameDev Tecnico** — a game development association involved in internal projects, external game jams, and international competitions.

Bachelor's in Computer Science & Engineering *Sep 2017 – Jun 2020*

University of Aveiro | GPA: 18.0 / 20.0

Highest grade average in the cohort and among the highest across the university. Worked as a paid **Undergraduate Researcher** developing a web-based data management BI platform. Served as a **board member** of **AETTUA**, an engineering student association focused on organizing university events and job fairs in close collaboration with external company partners.

PERSONAL PROJECTS

Full portfolio available at **https://heroufenix.github.io/**.

- **Blast Beat** (C++, Unreal Engine 5) - Solo PSX-Style arcade zombie shooter where the weapon fires to the beat of the song. Built entirely in one week, including all gameplay systems, enemy AI, upgrade mechanics and level design. Inspired by Call of Duty Zombies and BPM: Bullets Per Minute.
- **Incidence** (C++, Unreal Engine 5) - Online Multiplayer First Person Shooter built as a means to learn and start using Unreal Engine 5. Inspired by Ricochet.
- **Whitted Raytracer & GLSL Raytracer** (C++, OpenGL, GLSL) - Two-part rendering project implementing a CPU Whitted Raytracer at first and a GPU GLSL Progressive Raytracer. Covered features like BVH, soft shadows, depth of field, motion blur, fuzzy reflections and Russian Roulette optimisation.
- **Asteroids ML AI** (Unity, ML-Agents) - Trained a Reinforcement Learning agent to play a custom Asteroids recreation using Unity's ML-Agent framework, incorporating Curriculum Learning with ramping difficulty.

AWARDS, PUBLICATIONS & ACHIEVEMENTS

Published Article | IEEE International Symposium on Multimedia 2022 *Dec 2022*

Research paper accompanying Master's thesis, accepted and presented in Naples, Italy. The project - **Emotionally Expressive Motion Controller for Virtual Character Locomotion** - combined Machine Learning, Deep Reinforcement Learning, and Emotion Analysis (Laban Movement Analysis and the PAD model) to build a system capable of identifying and altering a virtual character's expressed emotion in real time, without additional animation data.

Best Art Direction | GameJobsLive GameParade GameJam *Apr 2022*

Awarded by industry professionals for the game **Felicitá**, which also ranked among the top audience-rated entries out of 45+ submissions.

Academic Excellence Award | University of Lisbon *May 2022*

Granted for outstanding academic performance and highest grade average in the MSc programme.

Game Developer | GameDev Técnico *Mar 2021 – Jul 2023*

Represented **GameDev Tecnico** - a student game development association - in international game jams and competitions, working as programmer and designer on award-winning projects. Selected to represent the association at **Nordic Game Jam 2022** in **Copenhagen**, with travel covered by the association. At the jam, collaborated with **LEGO** using their **proprietary in-house development kit** to produce a game in 48 hours.

Undergraduate Researcher | University of Lisbon & University of Aveiro *Dec 2019 – Sep 2021*

Revitalized a web-based data management platform wrapping Pentaho Kettle; developed supporting web platforms.